H-B-118E
October 4, 1976
SUPERSEDING
Fed. Spec. H B-118D
July 6, 1973

FEDERAL SPECIFICATION

BRUSHES: ARTISTS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers artists' brushes used for water and oil coloring, lettering, sign writing, and those in lithographic processes.

1.2 Classification.

1.2.1 Types, classes, styles, and sizes. Artists' brushes shall be of the following types, classes, styles, and sizes, as specified (see 6.2):

Types		Intended use
I	-	For water colors.
II	-	For oil and water colors.
III	-	Oils, varnishes, and lacquers (lettering and sign writing).
IV	-	Lithographic processes.

Classes		Materials
1	-	Squirrel tail hair.
2	-	Pure red sable hair.
3	-	Ox ear hair.

Styles		Shapes
A	-	Round, single taper point.
B	-	Flat, chisel edge.
C	-	Flat, square edge.
D	-	Round, square edge.

Sizes

(See tables I through V).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-B-838	-	Butyl Acetate, Normal (for Use in Organic Coatings).
TT-M-261	-	Methyl Ethyl Ketone, Technical.
PPP-B-1880	-	Brushes, Preparation for Delivery of.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Chemical Society (ACS) Specifications.

Reagent Chemicals (Fifth Edition)

Butyl Alcohol
Toluene

(Application for copies should be addressed to the American Chemical Society, 1155 Sixteenth Street, NW, Washington, DC 20036.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive and are not meant to preclude the purchase of brushes otherwise conforming to the specification.

3.2 Design. Brushes, as specified herein, shall consist of a handle with hair held in place by means of a ferrule and setting compound. Brushes shall be similar to figures 1 through 5 and shall conform to the dimensional requirements of tables I through V.

3.3 Material. Material shall be as specified herein.

3.3.1 Hair. Hair shall be of the best quality commercially available and as specified in 1.2 for each type of brush. All brush hair specified herein is commonly known by its commercial designation.

3.3.2 Ferrule. Ferrules at manufacturer's option shall be fabricated of seamless nonferrous metals or of nonferrous metals with a lockseam side overlap. Lengths, widths, shapes, and thickness of the ferrules shall be that used in standard manufacturer practice unless otherwise specified herein. Ferrules shall be fastened to the handle with a snug fit so that there is no play between the handle and the ferrule.

3.3.2.1 Finish. When tested as described in 4.4.2, any organic finish on the ferrule shall show no wrinkling, blistering, or loosening from the metal. After the brush has been removed from the solvent and air dried for 2 hours, any softening, dulling, or discoloration of the finish that was present at the time of removal of the brush from the solvent, shall have completely disappeared.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

TABLE I. MANUFACTURING REQUIREMENTS

3.3.3 Handle. Handles shall be of close-grained hardwood, lathe-shaped or turned similar to illustrations shown in figures 1 through 5. Handles shall be smoothly finished with a minimum of three dipped coats of varnish, lacquer or enamel. The coating shall not wrinkle, blister, or loosen when tested as specified in 4.4.3, or soften or dull when air-dried for 2 hours as specified in 4.4.2.

3.3.4 Setting compounds. The setting compound shall be vulcanized rubber, brushmaker's cement, or other suitable material which will hold the hair securely. The setting compound shall not injure the hair, shall not become brittle, and shall not extend beyond the edge of the ferrule. The setting compound shall be insoluble in lacquer, paint, varnish, water, thinners, and solvents.

3.3.5 Finished brushes. The finished brushes shall be free of loose hair, properly trimmed, or any other imperfections which may impair their durability and serviceability.

3.4 Marking. The handle of each brush shall be stamped or otherwise permanently marked with the manufacturer's name or trademark and the National Stock Number.

3.5 Type I, water colors.

3.5.1 Class 1, style A. Brushes shall be furnished in the style A design in class 1. The hair shall be squirrel tail hair for class 1, thoroughly sterilized and set to a single tapered point. The brushes shall be suitable for use with water colors. Type I class 1, style A brushes shall be similar to figure 1 and shall conform to the applicable dimensions of table I.

TABLE I. Minimum dimensional requirements for type I, class 1, style A brushes

Ferrule diameter inside at brush end (inch)	Length of hair outside of ferrule (inches)	Depth of ferrule (inches)	Hold of hair in ferrule (inch)	Length of handle (inches)
3/128	1/2	1-3/16	3/8	5
6/128	3/8	1	5/16	5
10/128	1/2	1-1/16	3/8	5
15/128	11/16	1-1/8	9/16	5
24/128	13/16	1-7/16	11/16	5
32/128	1-1/16	1-1/2	11/16	5
38/128	1-1/8	1-1/2	3/4	5
40/128	1-3/16	1-1/2	3/4	5

3.6 Type II, oil and water colors.

3.6.1 Class 2, style A. Type II brushes shall be furnished in the style A design in class 2. The hair shall be pure red sable for class 2, thoroughly sterilized and set to a single tapered point. The brushes shall be suitable for use with oils or with water colors, shall be similar to figure 2, and shall conform to the applicable dimensions of table II.

TABLE II. Minimum dimensional requirements for type II, class 2, style A brushes

Ferrule diameter inside at brush end (inch)	Length of hair outside of ferrule (inch)	Depth of ferrule (inches)	Hold of hair in ferrule (inch)	Length of handle (inches) 1/
6/128	5/16	1	5/16	9
8/128	7/16	1	5/16	9
10/128	1/2	1-1/16	1/2	9
13/128	9/16	1-1/8	5/8	9
15/128	5/8	1-1/8	5/8	9
18/128	11/16	1-1/4	5/8	9
26/128	13/16	1-7/16	5/8	9
30/128	7/8	1-1/2	5/8	9
34/128	1	1-1/2	5/8	9

1/ Unless otherwise specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

3.7 Type III, oils, varnishes, and lacquers (lettering and sign writing).

3.7.1 Class 3, style B. The hair in type III, class 3, style B, brushes shall be good quality, natural color, ox ear hair with natural points preserved and reversed hair removed. Hair shall be thoroughly sterilized and set to a flat, chisel edge. The brushes shall be suitable for use with oils, varnishes, and lacquers; be similar to figure 3, and shall conform to the applicable dimensions of table III.

TABLE III. Minimum dimensional requirements for type III, class 3, style B brushes

Brush width (inches)	Length of hair outside ferrule, minimum (inches)	Length of hair inside ferrule, minimum (inch)
1/8	11/16	1/2
1/4	3/4	1/2
3/8	13/16	1/2
1/2	15/16	1/2
5/8	1	1/2
3/4	1-1/16	1/2
7/8	1-3/16	1/2
1	1-1/4	1/2
1-1/4	1-5/16	1/2
1-1/2	1-3/8	1/2

3.7.2 Class 3, style D. The hair in type III, class 3, style D brushes shall be of the best quality ox ear hair having good life and spring and thoroughly cleaned and washed. The hair stock shall have good points, be free from turned hairs, and shall be set to a round, square, edge. The brush shall be suitable for use with oils, varnishes, and lacquers. Type III, class 3, style D brushes shall be similar to figure 4 and shall conform to the applicable dimensions of table IV.

TABLE IV. Minimum dimensional requirements for type III, class 3, style D brushes

Outside diameter of ferrule at same end (inch)	Length of hair outside ferrule (inches)	Depth of ferrule (inches)	Length of handle, including ferrule (inches)
1/16	3/4	1	7
3/32	7/8	1-1/16	7
1/8	1	1-1/16	7
5/32	1-3/8	1-1/8	7

3.8 Type IV, lithographic processes.

3.8.1 Class 1, style C. The hair in type IV, style C brushes shall be squirrel tail hair for class 1, thoroughly baked and set to a flat, square edge. The brushes shall be suitable for use in lithographic processes, be similar to figure 5, and conform to the applicable dimensions of table V.

TABLE V. Minimum dimensional requirements for type V, class 1, style C brushes

Ferrule width inside at brush end (inch)	Ferrule thickness outside at brush end (inch)	Length of hair outside ferrule (inch)	Depth of ferrule (inch)	Hold of hair in ferrule (inch)	Length of handle (inch)
1/2	5/32	7/8	1-1/2	1/2	5-3/4
1	7/32	1	1-3/4	1/2	5-3/4

3.9 Workmanship. The workmanship in the finished brush shall be equal to the minimum quality level generally accepted as good commercial practice in industry.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Sample units shall be assembled into lots as specified in paragraph 5 of MIL-STD-105. A sample unit shall be one brush for the purpose of visual inspection and testing.

4.2.2 Sampling for visual and dimensional examination. A random sample of brushes shall be selected from each lot offered for inspection in accordance with MIL-STD-105 at inspection level I. The acceptance quality level shall be equal to 4.0 percent defective.

4.2.3 Sampling for tests. A random sample of brushes shall be selected from each lot in accordance with MIL-STD-105 at inspection level S-4. The AQL shall be 2.5 percent defective.

4.2.3.1 Testing of sample brushes for lot acceptance. Each of the sample brushes selected in accordance with 4.2 shall be tested in accordance with 4.4.1 through 4.4.4 to verify compliance with this specification. Any sample brush which does not conform to these characteristics shall be cause for rejection and if the number of non-conforming brushes in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by that sample.

4.3 Examination. Each of the sample brushes selected in accordance with 4.2 shall be examined to verify compliance with this specification. Any brush in the sample containing one or more defects shall be cause for rejection, and if the number of defective brushes in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.4 Test procedures. Unless otherwise specified, standard testing conditions are 23 ± 1°C (73 ± 2°F) and a relative humidity of 50 ± 5 percent.

4.4.1 Temperature test. Each of the sample brushes selected in accordance with 4.2 shall be placed in a cabinet in which the temperature shall be raised to 155°F and held at this level for 4 hours, after which the brush shall be examined for evidence of loosening of hair or other damage. If none is apparent, the brush shall be replaced in the cabinet, the temperature of which shall then be lowered to minus 20°F, and maintained for 1 day, after which the brush shall again be examined for evidence of loosening of hair or other damage. Evidence of loosening of hair or evidence of any other damage to the brush shall constitute failure of this test.

4.4.2 Solubility tests. A solvent shall be prepared from components to the applicable specifications (see 2.1) and in the proportions shown below.

	Percent by volume
Normal Butyl Acetate (TT-B-838)	15 ± 1
Normal Butyl Alcohol (ACS Reagent grade)	15 ± 1
Methyl Ethyl Ketone (TT-M-261)	10 ± 1
Toluene (ACS Reagent grade)	Remainder

Each sample brush shall be suspended so that the hair and approximately one-half of the ferrule are immersed in the above solvent for 1 hour at standard laboratory conditions. The brush shall then be removed and the setting compound and finish on the ferrule examined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

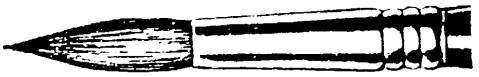


Figure 1 — For water colors, single tapered point.

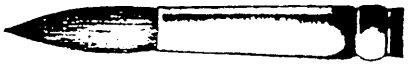


Figure 2 — For oil and water colors single tapered point.



Figure 3 — Ox ear hair, sign writer, flat chisel edge.

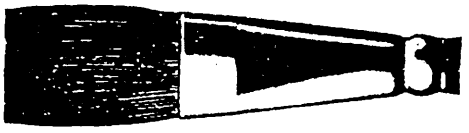


Figure 4 — Sable, lettering, round, square edge.



Figure 5 — Squirrel tail hair, lithographic processed, flat, chisel edge.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.

4.4.3 Setting compound. At the time of removal of the brush from the liquid, the setting compound shall be examined for excessive softening, loosening of the hair and evidence of solubility in the solvent. Evidence of solubility of the setting compound will be indicated by a change in the color of the solvent. The brush shall then be air-dried for a period of 24 hours at standard laboratory conditions, the ferrule removed, and the setting compound examined for evidence of deterioration or loosening of the hair or bristles.

4.4.4 Organic finish. At the time of removal of the brush from the liquid, the organic finish on the ferrule shall be examined for wrinkling, blistering, or loosening from the metal. The brush shall then be air-dried for 2 hours at standard laboratory conditions and the finish examined for any softening, dulling, or discoloration.

4.5 Inspection for preparation for delivery. The preservation, packaging, packing, and marking shall be examined for compliance with section 5 of this specification. The examination shall be conducted as specified in PPP-B-1880.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. The preservation (when applicable) packaging, packing, and marking shall be in accordance with PPP-B-1880. The level of packaging shall be A, B, or C, the level of packing shall be A, B, or C, and the marking shall be civil or military as specified (see 6.2).

6. NOTES

6.1 Intended use. The brushes covered by this specification are intended to be used, where applicable, for water and oil coloring, lettering, and sign writing, and in lithographic processing.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type, class, style, and size required (see 1.2, 3.5, 3.6, 3.7, 3.8).
(c) Level of packaging and packing required (see 5.1).
(d) Whether civil or military marking is required (see 5.1).

6.3 Supersession data. The following changes have been made.

H-b-118D	H-B-188E
Type I, class 1, style A	No change
Type II, class 3, style A	Type II, class 2, style A
Type III, class 5, style B	Type III, class 3, style B
Type III, class 5, style D	Type III, class 3, style D
Type V, class 1, style C	Type IV, class 1, style C

No changes in size designations.

CIVIL AGENCIES COORDINATING ACTIVITIES.

COMMERCE - NBS
CPS
DC GOVT. - DCG
GPO
GSA - FSS
POSTAL - POS
VA - DMS

PREPARING ACTIVITY: GSA - FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:06Z
Check the source to verify that this is the current version before use.